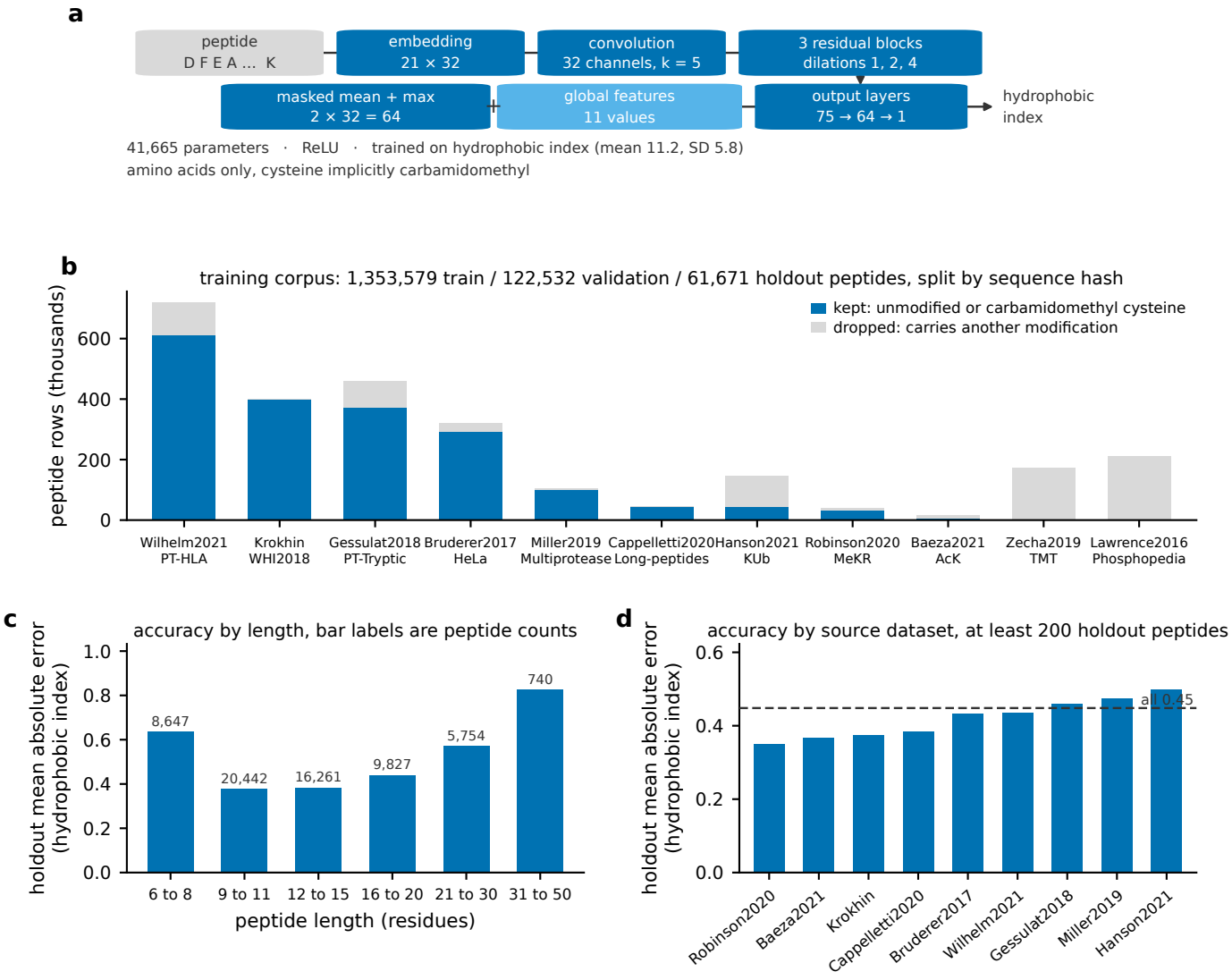

a
peptide
D F E A … K
embedding
21 × 32
convolution
32 channels, k = 5
3 residual blocks
dilations 1, 2, 4
masked mean + max
2 × 32 = 64
global features
11 values
output layers
75 → 64 → 1
hydrophobic index
41,665 parameters · ReLU · trained on hydrophobic index (mean 11.2, SD 5.8)
amino acids only, cysteine implicitly carbamidomethyl
b
training corpus: 1,353,579 train / 122,532 validation / 61,671 holdout peptides, split by sequence hash
kept: unmodified or carbamidomethyl cysteine
dropped: carries another modification
peptide rows (thousands)
0
200
400
600
Wilhelm2021 PT-HLA
Krokhin WHI2018
Gessulat2018 PT-Tryptic
Bruderer2017 HeLa
Miller2019 Multiprotease
Cappelletti2020 Long-peptides
Hanson2021 KUb
Robinson2020 MeKR
Baeza2021 AcK
Zecha2019 TMT
Lawrence2016 Phosphopedia
c
accuracy by length, bar labels are peptide counts
holdout mean absolute error (hydrophobic index)
0.0
0.2
0.4
0.6
0.8
1.0
8,647
20,442
16,261
9,827
5,754
740
6 to 8
9 to 11
12 to 15
16 to 20
21 to 30
31 to 50
peptide length (residues)
d
accuracy by source dataset, at least 200 holdout peptides
holdout mean absolute error (hydrophobic index)
0.0
0.2
0.4
0.6
all 0.45
Robinson2020
Baeza2021
Krokhin
Cappelletti2020
Bruderer2017
Wilhelm2021
Gessulat2018
Miller2019
Hanson2021